AP Calculus AB

Concepts and practice English edition

First edition for the 2027 exam

This independently authored book supports high-school students learning calculus and teachers preparing them for AP Calculus AB. Learn through concise notes and worked examples, explain reasoning in unit practice, and then apply those skills under exam timing.

Component	Count
Units and worked examples	8 / 16
Prerequisite diagnostic MCQ	12
Unit practice MCQ and FRQ	48 / 16
Mock exam MCQ and FRQ	42 / 6
Total assessment questions	124

Answers are separated into the teacher guide and linked by question ID. Use the diagnostic to identify prerequisite skills; a single score should not determine a student's potential.

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Course sequence and classroom use

Unit	Focus	Suggested weeks
1	Limits and continuity	1-3
2	Derivative definition and basic rules	4-5
3	Composite implicit and inverse derivatives	6-7
4	Contextual applications of differentiation	8-9
5	Analytical applications of differentiation	10-12
6	Integration and accumulation of change	13-16
7	Differential equations	17-18
8	Applications of integration	19-21

This is a suggested 21-week sequence with two 90-minute meetings per week. Use week 22 for the mock exam and review. This pacing is an instructional suggestion, not an official requirement. Add algebra or trigonometry support as the diagnostic indicates.

Before class, read the notes and redo examples with solutions covered. During class, connect definitions, diagrams, formulas, and units. For homework, follow MCQ with FRQ and write one sentence explaining each error.

In free response, state theorem hypotheses, variable meanings, integral bounds, and units. Show an expression before a numerical answer. Calculator work still requires mathematical setup and interpretation.

Exam format and solution conventions

Part	Questions	Minutes	Calculator
I A	29	62	Not permitted
I B	13	38	Graphing calculator required
II A	2	30	Graphing calculator required
II B	4	60	Not permitted

The table reflects the official format effective May 2027. On the hybrid exam, students answer MCQ and view FRQ in Bluebook, then handwrite FRQ answers in paper booklets. This print book reproduces timing practice rather than the testing software.

Angles are in radians unless stated otherwise. Retain full precision during calculator work and round only final numerical answers to three decimals. Exact expressions are acceptable. This is the rounding convention for this practice book.

Weight MCQ and FRQ at 50% each. Each original FRQ here has three 3-point parts, totaling 9. These instructional rubrics are author-created and do not constitute official scoring guidelines or an AP 1-to-5 score conversion.

$$C = 50\frac{M}{42} + 50\frac{R}{54}$$

Here M is correct mock MCQ and R is total mock FRQ points. The result C is a classroom percentage metric.

Prerequisite diagnostic 1 to 3

Work without a calculator and identify prerequisite gaps

D01

What is the larger root of $x^2 - 5x + 6 = 0$?

A. 3

B. 6

C. 2

D. 5

Work

D02

For $x \neq 4$, simplify $(x^2 - 16)/(x - 4)$.

A. $x + 4$

B. $x^2 + 4$

C. $x - 4$

D. 4

Work

D03

Find $\sin(\pi/6)$.

A. 0

B. $\frac{1}{2}$

C. $\frac{\sqrt{3}}{2}$

D. 1

Work

Prerequisite diagnostic 4 to 6

D04

Find $\ln(e^3)$.

A. 0

B. 3

C. e^3

D. 1

Work

D05

If $f(x)=x^2+1$ and $g(x)=2x-3$, find $f(g(2))$.

A. 7

B. 10

C. 1

D. 2

Work

D06

Find the slope through $(1,3)$ and $(5,11)$.

A. 2

B. $\frac{1}{2}$

C. 8

D. 4

Work

Prerequisite diagnostic 7 to 9

D07

What is the real domain of $\ln(x-2)$?

A. $x \geq 2$

B. All real numbers

C. $x > 2$

D. $x < 2$

Work

D08

For $f(x) = 3x + 5$, find $f^{-1}(11)$.

A. 2

B. $\frac{16}{3}$

C. 38

D. $\frac{1}{11}$

Work

D09

For $x > 0$, simplify $x^{3/2} / x^{1/2}$.

A. x

B. $\sqrt{x}$

C. 1

D. x^2

Work

Prerequisite diagnostic 10 to 12

D10

What is $\sin^2 x + \cos^2 x$?

A. 1

B. $\sin(2x)$

C. 2

D. 0

Work

D11

Find the minimum of $x^2 - 6x + 11$.

A. 11

B. 3

C. 2

D. -6

Work

D12

Find the sum of the two solutions of $|x - 2| = 5$.

A. 10

B. -3

C. 4

D. 7

Work

Unit 1 Limits and continuity

Distinguish function values from limits and justify conclusions using continuity and theorem hypotheses.

Reading approaching values

A limit describes nearby outputs as the input approaches a point. The value at that point may be missing or different. A two-sided finite limit exists only when both one-sided limits agree. Distinguish open and filled points on graphs. A finite table suggests behavior but does not prove a limit.

$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = L$$

Algebra and the squeeze theorem

Try substitution first. The form $0/0$ calls for further analysis. Factoring and canceling preserve the expression near the point, not necessarily at the point. Conjugates help with radicals. Trigonometric limit identities use radians. The squeeze theorem applies when bounding functions approach the same value.

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

Continuity and discontinuities

Continuity requires a defined value, an existing limit, and equality of the two. At interval endpoints use the one-sided limit from within the interval. A removable hole can be repaired by redefining a value; jumps and infinite discontinuities cannot. Retain domain restrictions from an original rational denominator.

$$\lim_{x \to a} f(x) = f(a)$$

Infinity and intermediate values

Infinite limits describe unbounded behavior near vertical asymptotes. Limits at infinity describe end behavior. Compare leading terms of rational functions and check signs. The intermediate value theorem requires continuity on a closed interval and a target between endpoint values. It guarantees existence, not uniqueness.

$$f(a) < N < f(b) \implies f(c) = N \text{ for some } c \in (a, b)$$

Unit 1 Learning through examples

Example 1

Find $\lim_{x \to 4} (x^2 - 16)/(x - 4)$.

For $x \neq 4$, factor the numerator as $(x - 4)(x + 4)$. The nearby expression is $x + 4$, so the limit is 8. The original expression is undefined at 4, yet its limit exists.

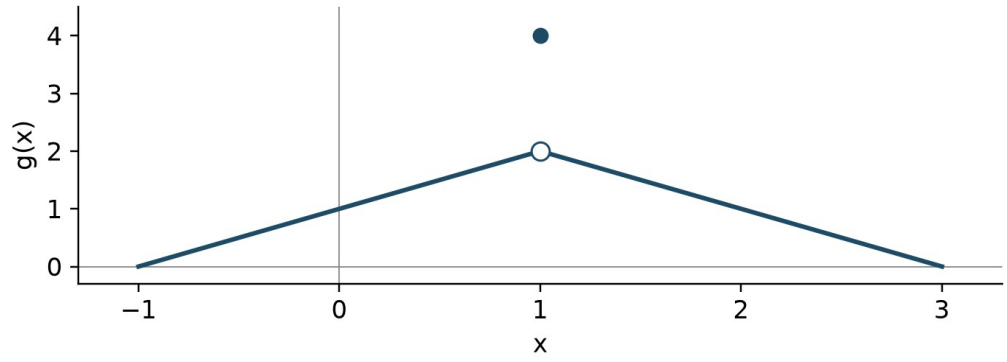
Error check Do not conclude that a zero denominator alone makes the limit nonexistent.

Example 2

Explain why $f(x) = x^3 + x - 5$ **has at least one zero in** $(1, 2)$.

The polynomial is continuous on $[1, 2]$. Its endpoint values are -3 and 5, which straddle zero. The intermediate value theorem gives a point $c \in (1, 2)$ with $f(c) = 0$.

Error check State continuity and endpoint signs, not only the theorem name.



An open point can separate a limit from the actual value.

Unit 1 Multiple choice practice 1 to 3

U1-M01

Evaluate $\lim_{x \to 3} (x^2 - 9)/(x - 3)$.

A. 3

B. 0

C. 6

D. 9

Work

U1-M02

Let $f(x) = 2 + x$ for $x < 0$ and $f(x) = 5 - x$ for $x \geq 0$. What is the two-sided limit at 0?

A. 5

B. $7/2$

C. Does not exist

D. 2

Work

U1-M03

For $x < 2$, $f(x) = k\,x + 1$; for $x \geq 2$, $f(x) = x^2 + 5$. Which k makes f continuous?

A. 4

B. 9

C. 2

D. 5

Work

Unit 1 Multiple choice practice 4 to 6

U1-M04

Evaluate $\lim_{x \to 0} x^2 \sin(1/x)$.

A. 2

B. 1

C. 0

D. -1

Work

U1-M05

Evaluate $\lim_{x \to \infty} (3x^2 - x)/(2x^2 + 7)$.

A. $\frac{2}{3}$

B. 3

C. 0

D. $\frac{3}{2}$

Work

U1-M06

A continuous f on $[0, 4]$ satisfies $f(0) = -2$ and $f(4) = 6$. Which must be true?

A. There is $c \in (0, 4)$ with $f(c) = 1$

B. There is exactly one zero

C. f is increasing

D. $f(2) = 2$

Work

U1-F01 Free response

Calculator not permitted

Define $f(x)=x^2+a$ for $x<1$, $f(1)=b$, and $f(x)=3x-1$ for $x>1$.

(a) Find the left and right limits at 1.

Work and justification

(b) Find a, b that make f continuous everywhere.

Work and justification

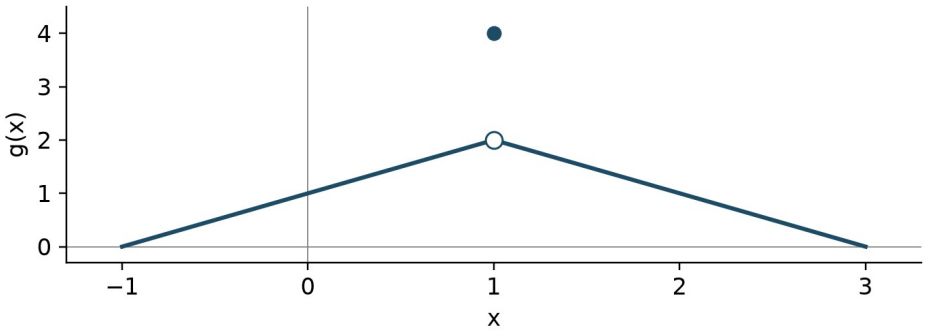
(c) Use part (b). Justify the existence of $c \in (0,2)$ with $f(c)=3$, and find one such value.

Work and justification

U1-F02 Free response

Calculator not permitted

The graph follows segments from $(-1,0)$ to $(1,2)$ to $(3,0)$, with a hole at $(1,2)$ and actual value $g(1)=4$.



(a) Find the two-sided limit and the function value at 1.

Work and justification

(b) Redefine one value to make g continuous at 1.

Work and justification

(c) For the original function, $g(0)=1$ and $g(1)=4$. Can IVT guarantee the value 3 on $[0,1]$? Does such a point actually exist?

Work and justification

Unit 2 Derivative definition and basic rules

Define instantaneous rate by a limit and apply basic differentiation rules accurately.

From average to instantaneous change

A secant slope is an average rate over an interval; a tangent slope is an instantaneous rate. The derivative is the limit of a difference quotient. A centered difference is often useful for estimates from tables, but distinguish an estimate from an exact derivative.

$$f'(a)=\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$$

Differentiability and continuity

Differentiability implies continuity. A continuous function can still have a corner, cusp, or vertical tangent and fail to have a finite derivative. At a piecewise junction first check continuity, then compare one-sided slopes. A tangent line needs both a function value and a derivative.

$$y=f(a)+f'(a)(x-a)$$

Basic derivatives

Combine the power, constant-multiple, and sum rules. Use the basic exponential, logarithmic, and trigonometric derivatives. The real domain of $\ln x$ is $x>0$, while the derivative of $\ln|x|$ is $1/x$ for $x\neq 0$. Trigonometric derivatives assume radian measure.

$$(e^x)'=e^x,(\sin x)'=\cos x,(\cos x)'=-\sin x$$

Products and quotients

A product derivative has two terms. A quotient derivative divides by the denominator squared and requires the correct subtraction order. Decide whether expanding or simplifying first is easier. Derivatives of tangent, cotangent, secant, and cosecant follow from sine and cosine ratios.

$$(uv)'=u'v+uv',\left(\frac{u}{v}\right)'=\frac{u'v-uv'}{v^2}$$

Unit 2 Learning through examples

Example 1

Use the definition to find $f'(3)$ for $f(x)=x^2$.

The quotient is $((3+h)^2-9)/h=6+h$. Taking $h \to 0$ gives $f'(3)=6$. The tangent line is $y-9=6(x-3)$.

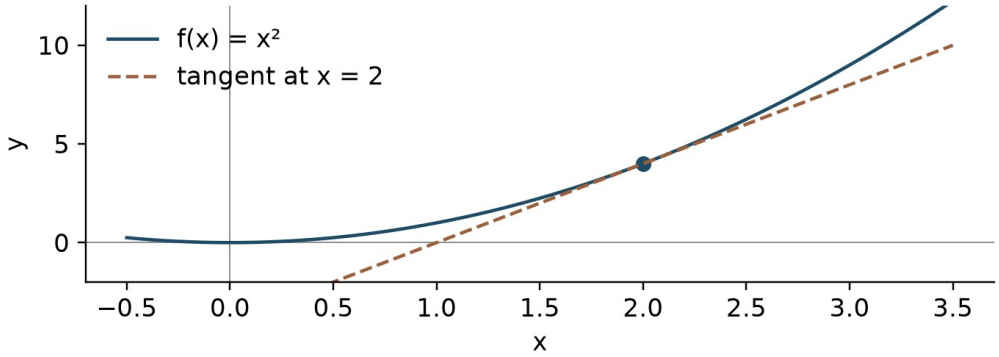
Error check Parenthesize the entire numerator of the difference quotient.

Example 2

Differentiate $f(x)=x^2e^x$.

The product rule gives $f'(x)=2xe^x+x^2e^x=e^x(x^2+2x)$. Multiplying the two derivatives is not the product rule.

Error check Include both terms in the product rule.



A tangent slope gives the instantaneous rate.

Unit 2 Multiple choice practice 1 to 3

U2-M01

Evaluate $\lim_{h \to 0} \left((2+h)^3 - 8 \right) / h$.

A. 4

B. 12

C. 6

D. 8

Work

U2-M02

If $f(x) = x^2 \sin x$, find $f'(\pi/2)$.

A. 0

B. 2

C. π

D. $\frac{\pi^2}{4}$

Work

U2-M03

For $f(x) = (x^2+1)/(x+1)$, find $f'(1)$.

A. 1

B. $\frac{1}{2}$

C. 2

D. 0

Work

Unit 2 Multiple choice practice 4 to 6

U2-M04

For $f(x)=\tan x$, find $f'(\pi/4)$.

A. $\sqrt{2}$

B. 1

C. 2

D. 0

Work

U2-M05

Which describes $f(x)=|x-2|$ at $x=2$?

A. Continuous but not differentiable

B. Its derivative is 0

C. Continuous and differentiable

D. Discontinuous

Work

U2-M06

The values of f at $x=1.8, 2, 2.2$ are $4.1, 4.8, 5.7$. Estimate $f'(2)$ using the outer two points.

A. 4

B. 0.8

C. 8

D. 2

Work

U2-F01 Free response

Calculator not permitted

Let $f(x) = x^3 - 2x$.

(a) Expand the difference quotient to find $f'(-1)$ from the definition.

Work and justification

(b) Find the tangent line at $x = -1$.

Work and justification

(c) Find $f''(-1)$ and explain whether it is the tangent slope.

Work and justification

U2-F02 Free response

Calculator not permitted

$P(t)$ is plant height in centimeters, where t is weeks. Assume P is differentiable and use the table.

t	P(t)
0	2
1	3.5
2	6
3	10

(a) Use $t = 1, 3$ to estimate $P'(2)$ and give units.

Work and justification

(b) Find the average height rate on $[0, 3]$.

Work and justification

(c) Use the estimated slope in (a) to approximate $P(2.2)$.

Work and justification

Unit 3 Composite implicit and inverse derivatives

Read function structure and select chain, implicit, or inverse differentiation.

Inside and outside

For a composition, evaluate the outer derivative at the inner function and multiply by the inner derivative. With several layers include one derivative from each layer. Combine chain rules with product or quotient rules when needed. Naming the layers before calculating helps prevent omissions.

$$\frac{d}{dx}f(g(x))=f'(g(x))g'(x)$$

Implicit and higher derivatives

Because y depends on x, differentiating a term involving y produces a factor of y'. Isolate y' after differentiating. For a second derivative retain that dependence and use the product rule. If a slope formula has zero denominator, examine the original curve and numerator before classifying the tangent.

$$x^2+y^2=r^2 \Longrightarrow y'=-\frac{x}{y}\,(y\neq 0)$$

Slopes of inverse functions

An inverse derivative is the reciprocal of the original derivative at the corresponding original input. Do not confuse an inverse input with an original input. The formula requires a local inverse and a nonzero original derivative. The two graphs reflect across $y=x$.

$$(f^{-1})'(b)=\frac{1}{f'(a)} \text{ when } f(a)=b$$

Inverse trigonometry and notation

Use the derivatives of arcsine and arctangent and include a chain factor for a composition. In this context $\sin^{-1}x$ denotes an inverse function, not $1/\sin x$. A higher derivative comes from differentiating the preceding derivative as a function.

$$(\arcsin x)'=\frac{1}{\sqrt{1-x^2}},(\arctan x)'=\frac{1}{1+x^2}$$

Unit 3 Learning through examples

Example 1

Differentiate $y=(1+3x^2)^4$.

Differentiate the outer fourth power, then multiply by the inner derivative $6x$. Thus $y'=4(1+3x^2)^3(6x)=24x(1+3x^2)^3$.

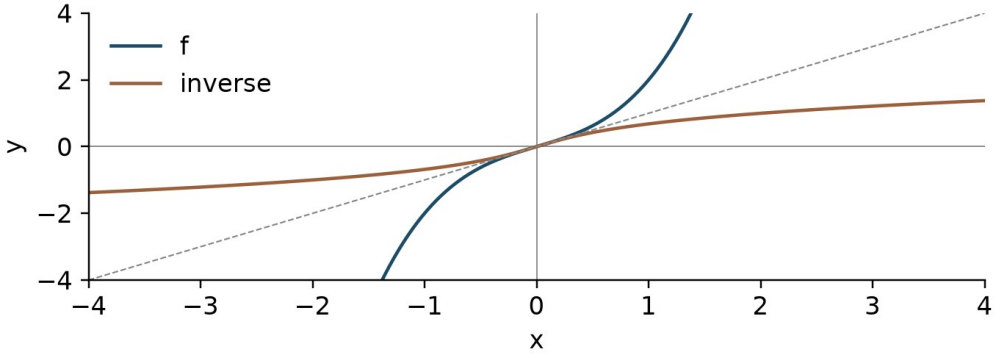
Error check Do not omit the inner derivative.

Example 2

If $f(2)=7$, $f'(2)=5$, and $g=f^{-1}$, find $g'(7)$.

Since $g(7)=2$, the derivative is $g'(7)=1/f'(2)=1/5$. The value $f'(7)$ is not needed.

Error check Find the corresponding point before taking a reciprocal.



A function and its inverse reflect across y=x.

Unit 3 Multiple choice practice 1 to 3

U3-M01

If $f(x)=(2x^3+1)^4$, find $f'(1)$.

A. 108

B. 648

C. 216

D. 72

Work

U3-M02

On $x^2+xy+y^2=7$, find dy/dx at $(1,2)$.

A. $-\frac{5}{4}$

B. $-\frac{1}{2}$

C. $\frac{4}{5}$

D. $-\frac{4}{5}$

Work

U3-M03

If $f(3)=8$, $f'(3)=4$, and g is its local inverse, find $g'(8)$.

A. 3

B. 4

C. $\frac{1}{4}$

D. $\frac{1}{8}$

Work

Unit 3 Multiple choice practice 4 to 6

U3-M04

For $f(x)=\arctan(3x)$, find $f'(0)$.

A. 3

B. $\frac{1}{3}$

C. 0

D. 1

Work

U3-M05

For $f(x)=\sin(2x)$, find $f''(\pi/4)$.

A. -2

B. 4

C. 0

D. -4

Work

U3-M06

Let g be the inverse of $f(x)=\ln x$. Find $g'(\ln 5)$.

A. 5

B. $\frac{1}{5}$

C. $\log(5)$

D. 1

Work

U3-F01 Free response

Calculator not permitted

Consider the point $(3, 4)$ on $x^2 + y^2 = 25$.

(a) Find dy/dx in terms of x, y and evaluate it at the point.

Work and justification

(b) Find the tangent line at the point.

Work and justification

(c) Find d^2y/dx^2 at the point.

Work and justification

U3-F02 Free response

Calculator not permitted

Let $f(x)=x^3+x$ and $g=f^{-1}$.

(a) Explain why the inverse exists and find $g(10)$.

Work and justification

(b) Find $g'(10)$.

Work and justification

(c) If $H(x)=g(5x)$, find $H'(2)$.

Work and justification

Unit 4 Contextual applications of differentiation

Interpret derivative units and apply derivatives to motion, related rates, and approximation.

Units and motion

Derivative units are output units divided by input units. Velocity is the derivative of position and acceleration is the derivative of velocity. Speed is absolute velocity, so a moving particle speeds up when velocity and acceleration have the same sign. Direction and speeding up are separate questions.

$v(t)=s'(t), a(t)=v'(t), \text{speed}=|v(t)|$

Related rates

Draw a diagram and identify quantities that vary with time. Write their relationship, differentiate with respect to time, then substitute values for the specified instant. In conical-tank problems use similar triangles to eliminate a variable first. Track length and volume units through the calculation.

$V=\frac{4}{3}\pi r^3 \Longrightarrow \frac{dV}{dt}=4\pi r^2\frac{dr}{dt}$

Linearization

Approximate a nearby function value with a tangent at an easy base point. Linearization is local. If the function is concave down between the base and target points, the tangent overestimates; if concave up, it underestimates. State the relevant concavity condition when judging error direction.

$f(a+\Delta x)\approx f(a)+f'(a)\Delta x$

L Hospital rule

First check for a $0/0$ or infinity-over-infinity form. Under the theorem's differentiability and denominator-derivative conditions, use the limit of the ratio of derivatives. This is different from differentiating the original quotient. Rewrite other indeterminate forms before applying the theorem.

$\frac{\lim f(x)}{g(x)}=\frac{\lim f'(x)}{g'(x)}$ when the hypotheses hold

Unit 4 Learning through examples

Example 1

A sphere has radius 2 cm and $dr/dt = 3$ cm/s. Find its volume rate.

Substitute into $dV/dt = 4\pi r^2(dr/dt)$ to obtain $4\pi(2)^2(3) = 48\pi$ cm³/s.

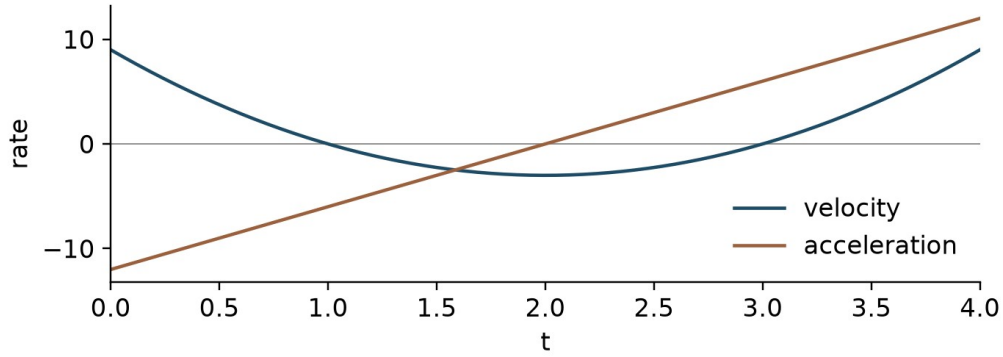
Error check Do not replace the changing radius with a constant before differentiating.

Example 2

Use linearization to estimate $\sqrt{9.12}$.

Since $f(9) = 3$ and $f'(9) = 1/6$, the estimate is $3 + (1/6)(0.12) = 3.02$. The square-root function is concave down, so this is an overestimate.

Error check Multiply the derivative by the input change.



Compare velocity and acceleration signs together.

Unit 4 Multiple choice practice 1 to 3

U4-M01

A particle has velocity $v(t)=3t^2-12t+9$. At $t=3/2$ it is:

A. Moving in the negative direction and slowing down

B. At rest

C. Moving in the positive direction and speeding up

D. Moving in the negative direction and speeding up

Work

U4-M02

When a sphere has radius 3 cm, $dr/dt=2$ cm/s. Find dV/dt in cm^3/s.

A. 18π

B. 24π

C. 36π

D. 72π

Work

U4-M03

Use linearization at $x=16$ to estimate $\sqrt{16.2}$.

A. 4

B. $\frac{41}{10}$

C. $\frac{161}{40}$

D. $\frac{81}{20}$

Work

Unit 4 Multiple choice practice 4 to 6

U4-M04

Evaluate $\lim_{x \to 0} \left(e^{2x} - 1\right) / x$.

A. 1

B. 0

C. 4

D. 2

Work

U4-M05

$W(t)$ is water volume in liters and t is hours. Interpret $W'(3) = -4$.

A. The rate itself decreases by 4 liters per hour each hour

B. At hour 3, water volume decreases at 4 liters per hour

C. Volume fell by 4 liters over the first 3 hours

D. There are 4 liters at hour 3

Work

U4-M06

A 10 m ladder has its foot 6 m from a wall, moving outward at 2 m/s. Find the top height rate in m/s.

A. $\frac{3}{2}$

B. -2

C. $-\frac{8}{3}$

D. $-\frac{3}{2}$

Work

U4-F01 Free response

Calculator not permitted

A particle has position $s(t) = t^3 - 6t^2 + 9t$ meters for $0 \le t \le 4$, with time in seconds.

(a) Find formulas for velocity and acceleration.

Work and justification

(b) At $t = 1.5$, justify the direction and whether the particle is speeding up.

Work and justification

(c) Use linearization at 1.5 to estimate $s(1.6)$.

Work and justification

U4-F02 Free response

Calculator not permitted

An inverted cone has height 8 cm and top radius 4 cm. Water enters at 3π cm^3/s. At a certain instant its water depth is 2 cm.

(a) Express water volume using only depth h.

Work and justification

(b) Find dh/dt at that instant.

Work and justification

(c) Linearly estimate the depth 0.1 second later. Justify whether it overestimates the actual depth.

Work and justification

Unit 5 Analytical applications of differentiation

Use theorem hypotheses and derivative signs to analyze graphs and extrema.

Mean and extreme values

The mean value theorem needs continuity on a closed interval and differentiability inside; it guarantees an instantaneous rate equal to the average rate. The extreme value theorem guarantees at least one absolute minimum and maximum for a continuous function on a closed bounded interval. Distinguish their hypotheses.

$f'(c)=\frac{f(b)-f(a)}{b-a}$ for some $c\in(a,b)$

Monotonicity and extrema

A critical number is in the function domain and has zero or undefined derivative. Positive derivative gives increase and negative derivative gives decrease. A change from positive to negative gives a local maximum; the reverse gives a minimum. For absolute extrema compare original function values at critical numbers and endpoints. A zero derivative alone is insufficient.

$f'(x)>0\Rightarrow f$ increases, $f'(x)<0\Rightarrow f$ decreases

Concavity and connected graphs

When $f''>0$, the first derivative increases and the graph is concave up. An inflection point requires a genuine concavity change; $f''=0$ only produces candidates. The second derivative test uses the sign of f'' where $f'=0$. If $f''=0$, that test is inconclusive. Do not mistake a derivative graph for a function graph.

$f'(c)=0,\ f''(c)>0\Rightarrow$ local minimum at c

Optimization and implicit analysis

In optimization separate the objective from constraints and identify the feasible domain. Reduce to one variable, then compare critical values with endpoints or boundary behavior. For implicit curves use y' and y'' to analyze tangents and concavity. Check that results meet the original length, time, or quantity restrictions.

$A(x)=x(P-2x)$ for a three-sided enclosure of fence length P

Unit 5 Learning through examples

Example 1

Find absolute extrema of $f(x)=x^3-3x$ on $[-2,2]$.

Critical numbers are -1 and 1. Values at endpoints and critical numbers are $-2, 2, -2, 2$, respectively. The absolute maximum is 2 and minimum is -2, each attained twice.

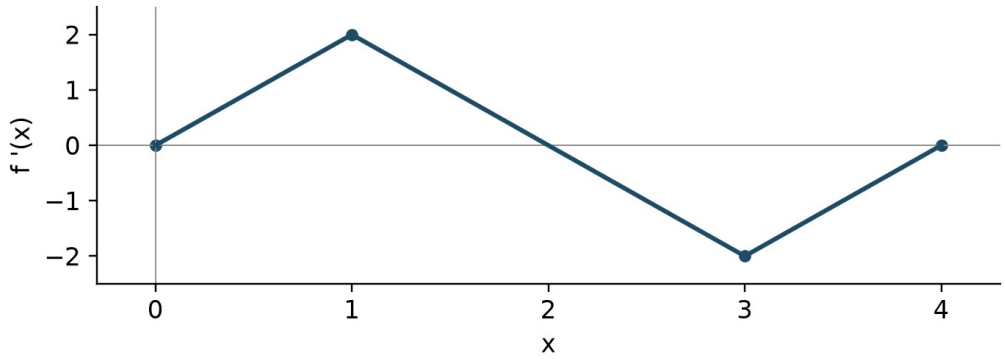
Error check Include endpoints and all locations attaining a tied extreme value.

Example 2

A river forms one side of a rectangle; 48 m of fence covers the other three. Find the maximum area.

Let the perpendicular side be x. The other side is $48-2x$, with $0<x<24$. Setting $A'=48-4x=0$ gives sides 12 and 24. The concave-down quadratic has maximum area 288 m^2.

Error check Do not count the river side as fenced.



This graph represents a derivative rather than the original function.

Unit 5 Multiple choice practice 1 to 3

U5-M01

For $f(x)=x^2$ on $[1,5]$, find the mean-value-theorem point c.

A. 4

B. 2

C. 3

D. 6

Work

U5-M02

Find the absolute maximum value of $f(x)=x^3-3x$ on $[-2,2]$.

A. -2

B. 0

C. 2

D. 6

Work

U5-M03

Which describes $x=0$ for $f(x)=x^3$?

A. A local minimum

B. Not a critical number

C. An inflection point but no local extremum

D. A local maximum

Work

Unit 5 Multiple choice practice 4 to 6

U5-M04

If $f'(x)=(x-1)(x-3)$, where is f increasing?

A. $(1,3)$

B. $(-\infty,1)$ and $(3,\infty)$

C. All real numbers

D. $(-\infty,3)$

Work

U5-M05

If $f''(x)=6x-12$ everywhere, find the inflection input.

A. 6

B. 2

C. 12

D. 0

Work

U5-M06

What is the maximum area in m^2 of a rectangle with perimeter 40 m?

A. 80

B. 200

C. 400

D. 100

Work

U5-F01 Free response

Calculator not permitted

Analyze $f(x) = x^3 - 6x^2 + 9x + 1$ on $[0, 4]$.

(a) Justify intervals of increase and decrease and locate interior local extrema.

Work and justification

(b) Find absolute extreme values and every location attaining them.

Work and justification

(c) Find the inflection point and explain the concavity change.

Work and justification

U5-F02 Free response

Calculator not permitted

Cut squares of side x cm from each corner of an 18 cm square sheet and fold an open-top box.

(a) Write the volume function and physically feasible domain.

Work and justification

(b) Justify the maximizing cut size and find the maximum volume.

Work and justification

(c) If equipment restricts cuts to $0 < x \leq 2$, what is the maximum volume?

Work and justification

Unit 6 Integration and accumulation of change

Interpret a definite integral as signed accumulation and use the fundamental theorem and substitution.

Riemann sums and numerical integration

Partition the interval and add width times representative height. Left, right, and midpoint sums differ in their sample points. Trapezoids average endpoint heights. With unequal widths multiply each width separately. A definite integral gives signed area, so regions below the axis contribute negatively.

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(a+i\,\Delta x)\,\Delta x$$

Accumulation and the fundamental theorem

For a continuous integrand, differentiating accumulation from a fixed lower endpoint to a variable upper endpoint returns the integrand. Include a chain factor for a composed upper endpoint. If both endpoints vary, write a difference of accumulations. Integrand signs control accumulation's increase and integrand slopes control its concavity.

$$\frac{d}{dx}\int_a^{g(x)} f(t)\,dt = f(g(x))\,g'(x)$$

Antiderivatives and integral properties

An antiderivative differentiates to the integrand; include C for an indefinite integral. A definite integral equals an antiderivative's endpoint difference. Reversing bounds changes the sign, and adjacent intervals add. Odd and even symmetry can simplify symmetric intervals. Check an antiderivative by differentiating it.

$$\int_a^b f(x)\,dx = F(b) - F(a) \text{ when } F' = f$$

Substitution and algebraic preparation

Try substitution when an inner expression appears with its derivative. For definite integrals change the bounds or return to the original variable before evaluating. Polynomial division and completing the square can reveal basic forms. Recognize arctangent and arcsine patterns. Integration by parts, partial fractions, and improper integrals are reserved for the BC extension.

$$\int f(g(x))\,g'(x)\,dx = F(g(x)) + C$$

Unit 6 Learning through examples

Example 1

Evaluate $\int_0^1 2x(x^2+1)^3\,dx$.

Set $u=x^2+1$ and $du=2x\,dx$; the new bounds are 1 and 2. Then $\int_1^2 u^3\,du=[u^4/4]_1^2=15/4$.

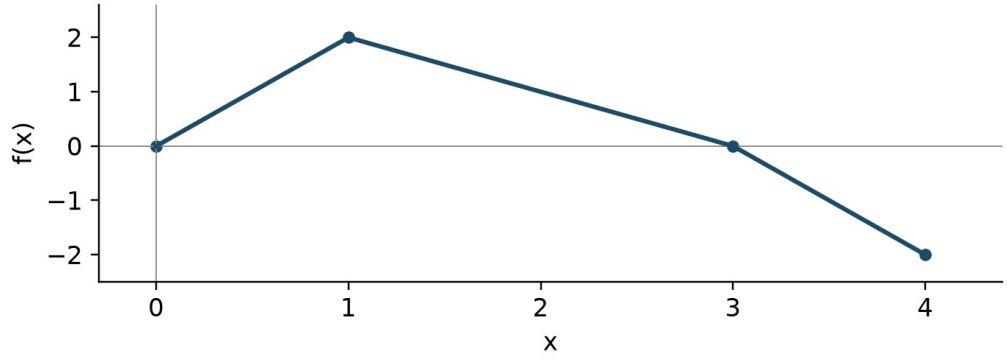
Error check Do not mix transformed variables with original bounds.

Example 2

If $F(x)=\int_1^{x^2}\cos t\,dt$, **find** $F'(x)$.

Evaluate the integrand at the upper endpoint and multiply by its derivative: $F'(x)=2x\cos(x^2)$.

Error check Do not omit the upper-endpoint derivative $2x$.



Areas below the axis contribute negatively to a signed integral.

Unit 6 Multiple choice practice 1 to 3

U6-M01

Evaluate $\int_0^2 (3x^2+1)\,dx$.

A. 14

B. 8

C. 6

D. 10

Work

U6-M02

If $F(x)=\int_0^{x^2} \sqrt{1+t}\,dt$, find $F'(2)$.

A. $4\sqrt{5}$

B. $2\sqrt{5}$

C. $4\sqrt{3}$

D. $\sqrt{5}$

Work

U6-M03

At $t=0,1,3$, the values of $r(t)$ are $2,4,8$. Estimate $\int_0^3 r(t)\,dt$ with two trapezoids.

A. 14

B. 18

C. 12

D. 15

Work

Unit 6 Multiple choice practice 4 to 5

U6-M04

Evaluate $\int_0^1 6x(1+3x^2)^2\,dx$.

A. 63

B. 7

C. 21

D. 16

Work

U6-M05

For $x>0$, which is an antiderivative of $(x^2+1)/x$?

A. $2x-\frac{1}{x^2}$

B. $\log(x^2+1)$

C. $x^2+\log(x)$

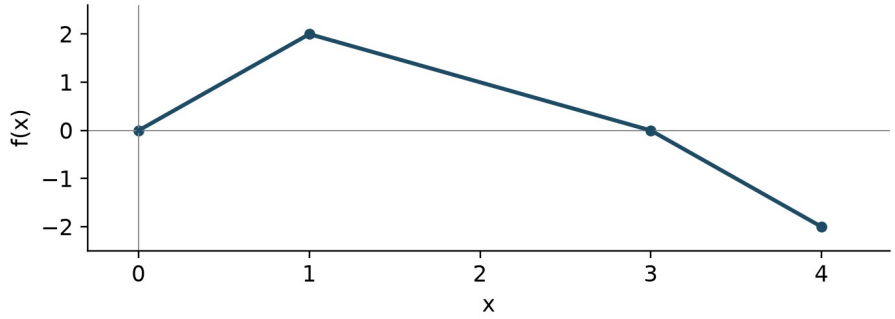
D. $\frac{x^2}{2}+\log(x)$

Work

Unit 6 Multiple choice practice 6 to 6

U6-M06

The graph of f consists of line segments. If $F(x)=\int_0^x f(t)\,dt$, find $F(3)$.



A. 6

B. 4

C. 3

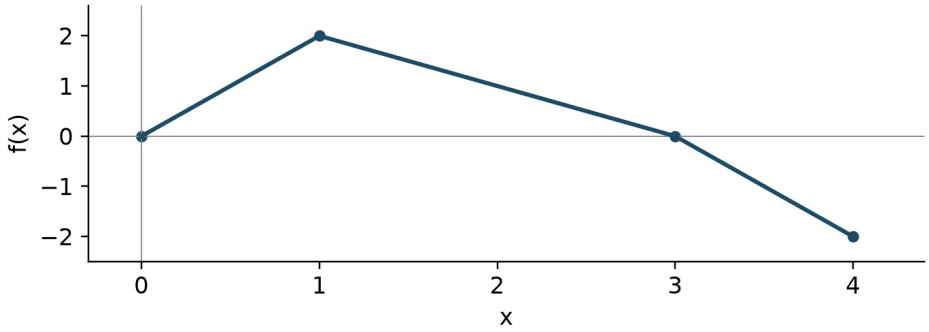
D. 2

Work

U6-F01 Free response

Calculator not permitted

The graph shows f. Define $F(x)=1+\int_0^x f(t)\,dt$ on $0\le x\le 4$.



(a) Find $F(3)$ and $F(4)$.

Work and justification

(b) Find $F'(2)$ and $F''(2)$.

Work and justification

(c) Justify the absolute maximum of F and its location.

Work and justification

U6-F02 Free response

Calculator not permitted

$r(t)$ is inflow in liters per minute. Initially a tank contains 20 liters. Assume r is continuous and water only enters.

t	r(t)
0	3
2	5
5	4
6	7

(a) Estimate $\int_0^6 r(t)\,dt$ with a left Riemann sum on the three tabulated intervals.

Work and justification

(b) Use trapezoids over the same intervals.

Work and justification

(c) Use (b) to estimate volume at 6 and average inflow on $[0,6]$.

Work and justification

Unit 7 Differential equations

Interpret slope fields and use separation with initial conditions.

Equations and slope fields

A differential equation relates a function to its rate of change. Each slope-field segment gives the derivative at its (x, y) location. If the rate depends only on y, slopes agree along horizontal rows; if only on x, along vertical columns. An equation may have infinitely many solutions.

$$\frac{dy}{dx} = F(x, y)$$

Verification and equilibrium

Verify a candidate by substituting both the function and its derivative into the original equation, then check the initial condition separately. An equilibrium solution is constant. When dividing by a factor during separation, check whether zero values of that factor yield equilibrium solutions that would otherwise be lost.

$$y' = k(y - M) \Longrightarrow y = M \text{ is an equilibrium}$$

Separation and initial values

For $dy/dx = g(x)h(y)$, separate as $dy/h(y) = g(x)dx$ on an interval where division is valid. Integrate and determine the constant from the initial condition. When solving with logarithms or square roots retain the appropriate branch and domain. Differentiate the result to check the equation and initial value.

$$\int \frac{1}{h(y)} dy = \int g(x) dx$$

Exponential models and cooling

A rate proportional to the quantity produces an exponential model. If the rate is proportional to the difference from a baseline, model that difference first. Find half-lives or doubling times by substituting into the exponential equation. Euler's method and detailed logistic modeling belong to the later BC extension.

$$y' = k(y - M) \Longrightarrow y = M + Ce^{kx}$$

Unit 7 Learning through examples

Example 1

Solve $y' = 2xy$ with $y(0) = 3$.

Separate to get $\ln|y| = x^2 + C$, hence $y = Ae^{x^2}$. The initial condition gives $A = 3$. Differentiating $3e^{x^2}$ returns $2xy$.

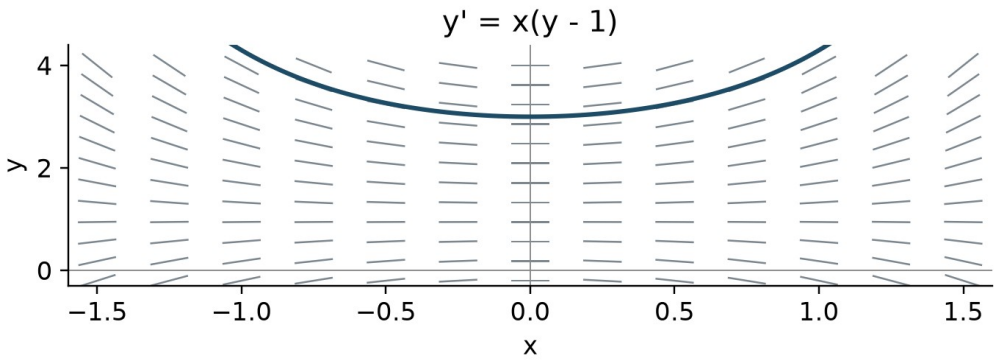
Error check Determine the integration constant from the initial value.

Example 2

Solve $T' = -0.2(T - 20)$ with $T(0) = 70$.

The excess temperature is $T - 20 = Ce^{-0.2t}$. The initial condition gives $C = 50$, so $T = 20 + 50e^{-0.2t}$, approaching 20 over time.

Error check Exponentially decay the excess temperature, not the entire initial temperature.



Short segments show local slopes. The curve solves the displayed equation with y(0)=3.

Unit 7 Multiple choice practice 1 to 3

U7-M01

If $y' = 3y$ and $y(0) = 2$, find $y(\ln 2)$.

A. 8

B. 16

C. 4

D. 6

Work

U7-M02

For $y' = x - y$, what is the slope at $(1, 3)$?

A. -2

B. -4

C. 4

D. 2

Work

U7-M03

For $y' = x / y$ with $y(0) = 2$, find $y(1)$.

A. $\sqrt{3}$

B. $-\sqrt{5}$

C. 5

D. $\sqrt{5}$

Work

Unit 7 Multiple choice practice 4 to 6

U7-M04

Find the equilibrium solution of $y' = -2(y-7)$.

A. -7

B. 7

C. 0

D. 2

Work

U7-M05

Which equation is satisfied by $y = Ce^{-2x}$ for every constant C?

A. $y' = y^2$

B. $y' = 2y$

C. $y' = -2y$

D. $y' = -2x$

Work

U7-M06

If $T' = -\frac{1}{2}(T-20)$ and $T(0) = 60$, find $T(2)$.

A. $\frac{40}{e} + 20$

B. $\frac{40}{e}$

C. $20 + \frac{60}{e}$

D. $\frac{60}{e}$

Work

U7-F01 Free response

Calculator not permitted

Consider $y' = x(y - 1)$ with $y(0) = 3$.

(a) Describe where the slope field has zero slope and find the initial tangent slope.

Work and justification

(b) Solve the initial-value problem.

Work and justification

(c) Determine the solution concavity on the real line.

Work and justification

U7-F02 Free response

Calculator not permitted

Temperature satisfies $T' = -k(T - 18)$ with $k > 0$, $T(0) = 82$, and $T(10) = 50$. Time is in minutes and temperature in degrees Celsius.

(a) Express T in terms of k, t.

Work and justification

(b) Find the exact value of k.

Work and justification

(c) Find the time when temperature reaches 26.

Work and justification

Unit 8 Applications of integration

Use integration for averages, accumulated quantities, area, and volume with appropriate units.

Average values and accumulated quantities

A function's average value is its integral divided by interval length; it is not average rate of change. Final amount equals initial amount plus net change. Integrate inflow minus outflow when both occur. To maximize an amount compare times when the net rate is zero with interval endpoints.

$$f_{avg} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

Displacement and distance

Integrating velocity gives displacement; integrating absolute velocity gives total distance. Find velocity zeros and split intervals when the sign changes. Add the initial position to recover position. Zero displacement can coexist with positive distance traveled.

$$\Delta s = \int_a^b v(t)\,dt, D = \int_a^b |v(t)|\,dt$$

Areas between curves

For vertical slices integrate top minus bottom with respect to x; for horizontal slices integrate right minus left with respect to y. Find all intersections and split where ordering changes. Distinguish signed integrals from geometric area. Check that the diagram and integral describe the same region.

$$A = \int_a^b (\text{top} - \text{bottom})\,dx$$

Cross sections and solids of revolution

Volume is the integral of cross-sectional area. A square has area side squared, an equilateral triangle has area $\sqrt{3}/4$ times side squared, and a semicircle with diameter d has area $\pi d^2/8$. Disk and washer radii are distances from the rotation axis. Redefine these distances for shifted horizontal or vertical axes.

$$V = \int_a^b A(x)\,dx, V = \pi\int_a^b (R^2 - r^2)\,dx$$

Unit 8 Learning through examples

Example 1

For $v(t)=t-2$ on $0\le t\le 4$, find displacement and total distance.

Displacement is $\int_0^4 (t-2)\,dt=0$. Split at $t=2$: distance is $-\int_0^2 (t-2)\,dt+\int_2^4 (t-2)\,dt=4$.

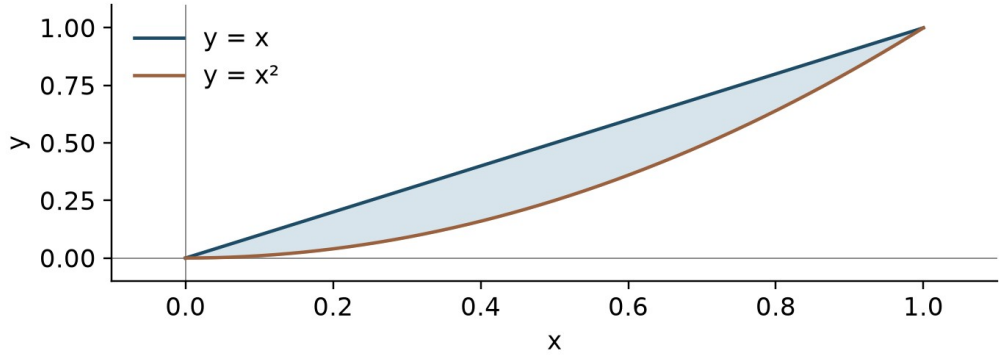
Error check Opposite directions do not cancel in total distance.

Example 2

Rotate the region between $y=x$ and $y=x^2$, $0\le x\le 1$, about the x-axis. Find its volume.

The outer radius is x and inner radius is x^2, so $V=\pi\int_0^1 (x^2-x^4)\,dx=2\pi/15$.

Error check Integrate the difference of squares, not the square of a difference.



The vertical slice length is top curve minus bottom curve.

Unit 8 Multiple choice practice 1 to 2

U8-M01

Find the average value of $f(x)=x^2$ on $[0,3]$.

A. 9

B. 6

C. $\frac{9}{2}$

D. 3

Work

U8-M02

A particle has $v(t)=t-1$ m/s for $0\le t\le 3$. Find total distance in meters.

A. $\frac{3}{2}$

B. 3

C. 2

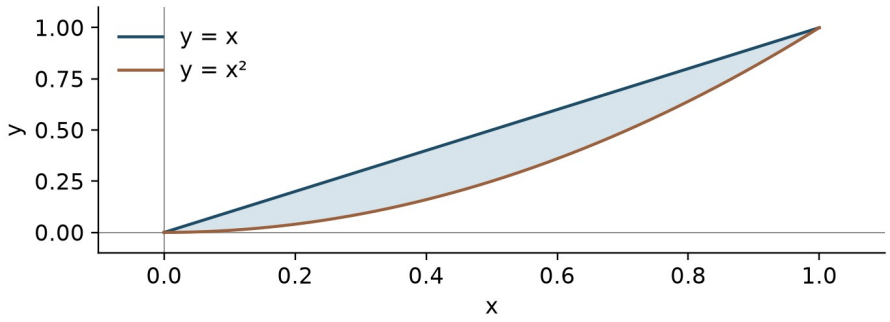
D. $\frac{5}{2}$

Work

Unit 8 Multiple choice practice 3 to 4

U8-M03

Find the area between $y = x$ and $y = x^2$ on $0 \le x \le 1$.



A. $\frac{2}{3}$

B. $\frac{1}{3}$

C. $\frac{1}{6}$

D. $\frac{1}{2}$

Work

U8-M04

Rotate the region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ about the x-axis. Find the volume.

A. 4π

B. $\frac{32\pi}{3}$

C. 8π

D. 16π

Work

Unit 8 Multiple choice practice 5 to 6

U8-M05

A solid has base between $y = x$ and $y = x^2$. Cross sections perpendicular to the x-axis are squares. Find its volume.

A. $\frac{1}{15}$

B. $\frac{1}{30}$

C. $\frac{1}{6}$

D. $\frac{\pi}{30}$

Work

U8-M06

Find the area enclosed by $x = y^2$ and $x = 2y$.

A. $\frac{8}{3}$

B. 4

C. $\frac{4}{3}$

D. $\frac{2}{3}$

Work

U8-F01 Free response

Calculator not permitted

A particle has $v(t)=(t-1)(t-3)$ m/s, $s(0)=2$ meters, and $0 \le t \le 4$.

(a) Find $s(4)$.

Work and justification

(b) Find total distance traveled.

Work and justification

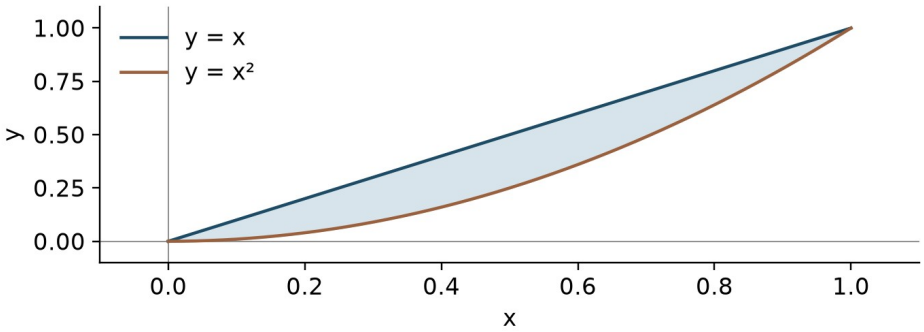
(c) Find the absolute maximum position and all times attaining it.

Work and justification

U8-F02 Free response

Calculator not permitted

Let R be the bounded region between $y = x$ and $y = x^2$.



(a) Find the intersection inputs and area.

Work and justification

(b) Find the volume when R revolves about the x-axis.

Work and justification

(c) Find the volume when R revolves about $y = 2$.

Work and justification

Mock exam I A 1 to 3

29 questions 62 minutes no calculator

E01

Evaluate $\lim_{x \to 4} \left(\sqrt{x+5}-3\right)/\left(x-4\right)$.

A. $\frac{1}{3}$

B. $\frac{1}{6}$

C. 0

D. 3

Work

E02

For $f\left(x\right)=e^{x}+x^{-2}$, find $f'\left(1\right)$.

A. e

B. $-2+e$

C. $-1+e$

D. $2+e$

Work

E03

Find the tangent slope at $\left(1,2\right)$ on $x\,y+y^{2}=6$.

A. $-\frac{2}{5}$

B. $-\frac{5}{2}$

C. $-\frac{1}{2}$

D. $\frac{2}{5}$

Work

Mock exam I A 4 to 6

E04

$T(t)$ is degrees Celsius and t is minutes. What does $T'(5)=3.2$ mean?

A. It rose 3.2 degrees over five minutes

B. The temperature rate increases by 3.2 each minute

C. The temperature at minute 5 is 3.2 degrees

D. At minute 5, temperature rises at 3.2 degrees per minute

Work

E05

For $f(x)=\ln x$ on $[1, e^2]$, find the MVT point c.

A. $\frac{2}{-1+e^2}$

B. $\frac{1}{2}+\frac{e^2}{2}$

C. e

D. $-\frac{1}{2}+\frac{e^2}{2}$

Work

E06

Evaluate $\int_{-1}^{1}(3x^2+2x)\,dx$.

A. 4

B. 0

C. 2

D. 6

Work

Mock exam I A 7 to 9

E07

If $y' = 2y$ and $y(0) = 5$, find $y(1)$.

A. 7

B. $5e^2$

C. 10

D. $5e$

Work

E08

Find the area bounded by $y = 2x - x^2$ and the x-axis.

A. $\frac{2}{3}$

B. $\frac{8}{3}$

C. 2

D. $\frac{4}{3}$

Work

E09

Evaluate $\lim_{x \to 2} 1/(x-2)^2$.

A. 0

B. ∞

C. $-\infty$

D. 1

Work

Mock exam I A 10 to 12

E10

At $x=1,2,3$, $f(x)=5,8,15$. Estimate $f'(2)$ using $x=1,3$.

A. 10

B. 3

C. 5

D. 7

Work

E11

For $f(x)=\arcsin(x/2)$, find $f'(1)$.

A. $\frac{2\sqrt{3}}{3}$

B. $\sqrt{3}$

C. $\frac{1}{2}$

D. $\frac{\sqrt{3}}{3}$

Work

E12

Evaluate $\lim_{x\to 0} \ln(1+3x)/\sin(2x)$.

A. 3

B. 0

C. $\frac{3}{2}$

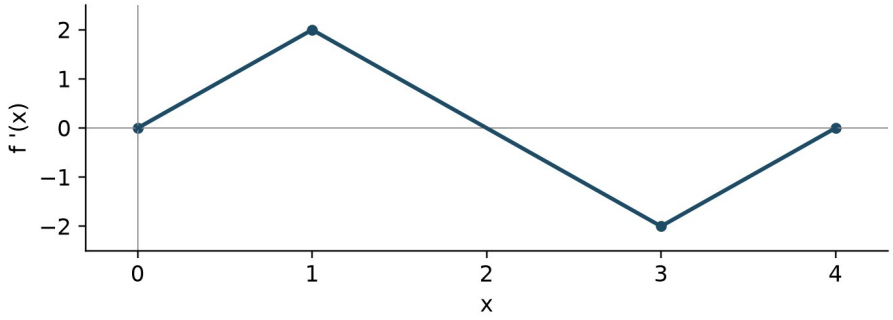
D. $\frac{2}{3}$

Work

Mock exam I A 13 to 14

E13

The graph shows f'. On which open intervals is f concave up?



A. $(1,3)$

B. $(0,2)$

C. $(2,4)$

D. $(0,1)$ and $(3,4)$

Work

E14

Evaluate $\lim_{n\to\infty}\sum_{i=1}^{n}(2/n)(1+2i/n)^2$.

A. 9

B. $\frac{8}{3}$

C. $\frac{26}{3}$

D. $\frac{13}{3}$

Work

Mock exam I A 15 to 17

E15

When separating $y' = x(y+4)$, one divides by $y+4$. Which constant solution must be checked separately?

A. 0

B. -4

C. $-x$

D. 4

Work

E16

Find the average value of $\sin x$ on $[0, \pi]$.

A. 1

B. 2

C. $\frac{\pi}{2}$

D. $\frac{2}{\pi}$

Work

E17

Which correctly justifies a zero of $p(x) = x^3 + 2x - 1$ in $(0,1)$?

A. Continuity and $p(0) = -1 < 0 < 2 = p(1)$ allow IVT

B. Because $p(0) + p(1) > 0$

C. Opposite endpoint signs suffice without continuity

D. Every cubic has a zero in $(0,1)$

Work

Mock exam I A 18 to 20

E18

For $f(x)=x\sin x$, find $f'(\pi)$.

A. π

B. -1

C. 0

D. $-\pi$

Work

E19

Water depth rises at 3 m/min in a cylinder of fixed radius 2 m. Find the volume rate in m^3/min.

A. 24π

B. 12π

C. 4π

D. 6π

Work

E20

Find the absolute minimum of $f(x)=x^4-4x^2$ on $[-2,2]$.

A. -4

B. 4

C. 0

D. -8

Work

Mock exam I A 21 to 23

E21

Evaluate $\int_0^1 1/(1+x^2)\,dx$.

A. $\frac{\pi}{4}$

B. $\frac{1}{2}$

C. $\frac{\pi}{2}$

D. $\log(2)$

Work

E22

Rotate the region bounded by $x=\sqrt{y}$, the y-axis, and $y=4$ about the y-axis. Find the volume.

A. 8π

B. 16π

C. $\frac{32\pi}{3}$

D. 4π

Work

E23

Evaluate $\lim_{x\to 0} \sin(5x)/(3x)$.

A. $\frac{3}{5}$

B. 1

C. $\frac{5}{3}$

D. 0

Work

Mock exam I A 24 to 26

E24

Let $f(x) = a\,x + b$ for $x < 1$ and $f(x) = x^2$ for $x \geq 1$. If f is differentiable at 1, find b.

A. 1

B. 0

C. 2

D. -1

Work

E25

What is the concavity of $y = \sqrt{9 - x^2}$ on $-3 < x < 3$?

A. Concave up throughout

B. Concavity changes at zero

C. Linear

D. Concave down throughout

Work

E26

Which is an antiderivative of $1/(x^2 - 2\,x + 5)$?

A. $atan(x - 1)$

B. $\log(x^2 - 2\,x + 5)$

C. $2\,atan\left(\frac{x}{2} - \frac{1}{2}\right)$

D. $\frac{atan\left(\frac{x}{2} - \frac{1}{2}\right)}{2}$

Work

Mock exam I A 27 to 29

E27

Cross sections are semicircles of diameter $2 - x$ for $0 \le x \le 2$. Find the volume.

A. $\frac{4\pi}{3}$

B. $\frac{2\pi}{3}$

C. $\frac{\pi}{3}$

D. $\frac{8}{3}$

Work

E28

If $f'(x) = x^2(x+1)$, which statement about local extrema is correct?

A. A local minimum at -1 and no extremum at 0

B. There is a maximum at -1

C. Both points are minima

D. Only 0 is a maximum

Work

E29

If $F(x) = \int_{x}^{2x} \cos(t^2)\,dt$, find $F'(0)$.

A. 0

B. 1

C. 2

D. -1

Work

Mock exam I B 1 to 3

13 questions 38 minutes graphing calculator required

E30

Find the zero of $p(x)=x^3+x-4$ in $[1,2]$ to three decimal places.

A. 1.812

B. 1.379

C. 1.587

D. 1.213

Work

E31

For $f(x)=x^2e^x$, find $f'(0.7)$.

A. 0.987

B. 3.806

C. 1.904

D. 2.819

Work

E32

Let $f(x)=x+e^x$ and $g=f^{-1}$. If $b=1.2+e^{1.2}$, find $g'(b)$.

A. 0.432

B. 0.231

C. 4.32

D. 0.301

Work

Mock exam I B 4 to 6

E33

If $v(t)=\sin(t^2)+0.2t$ m/s, find acceleration at $t=1.4$ in m/s^2.

A. 0.862

B. -0.862

C. -0.379

D. 1.205

Work

E34

Where does $f(x)=x^2+e^{-x}$ attain its absolute minimum on $[0,2]$?

A. 0.693

B. 1.0

C. 1.256

D. 0.352

Work

E35

Find the absolute maximum value of $f(x)=2\sin x-x^2/2$ on $[0,\pi]$.

A. 1.03

B. 2.0

C. 0.816

D. 1.184

Work

Mock exam I B 7 to 8

E36

Evaluate $\int_0^1 e^{-x^2}\,dx$ numerically.

A. 0.842

B. 0.632

C. 1.128

D. 0.747

Work

E37

For $F(x)=\int_0^{\ln x} e^{-t^2}\,dt$, find $F'(2)$.

A. 0.24

B. 0.309

C. 0.619

D. 0.347

Work

Mock exam I B 9 to 10

E38

Use three trapezoids from the table to estimate $\int_0^2 f(x)\,dx$.

x	f(x)
0	1
0.5	1.6
1.5	2.1
2	2.4

A. 3.45

B. 4.1

C. 3.8

D. 3.625

Work

E39

If $y' = 0.4(10 - y)$ and $y(0) = 2$, when does y first reach 8?

A. 3.466

B. 1.733

C. 2.5

D. 5.198

Work

Mock exam I B 11 to 13

E40

Find the area enclosed by $y = e^{-x}$, $y = x/2$, and the y-axis.

A. 0.574

B. 0.853

C. 0.608

D. 0.392

Work

E41

For $v(t) = \cos t - 0.2$ m/s, find total distance on $[0, 4]$ in meters.

A. 3.517

B. 2.969

C. 0.757

D. 1.557

Work

E42

If $T(t) = 20 + 80\,e^{-0.3t}$, find $T'(4)$.

A. 44.096

B. -7.229

C. -24.096

D. 7.229

Work

EF01 Free response

II A 2 questions total 30 minutes

Graphing calculator required

A tank initially contains 40 liters. For $0 \le t \le 4$ minutes, inflow is $R(t) = 12 + 3\sin(t^2/4)$ L/min and outflow is $D(t) = 8 + t$ L/min.

(a) Find the volume at $t = 4$ to three decimal places.

Work and justification

(b) Justify the time when water volume is greatest on $[0, 4]$.

Work and justification

(c) At $t = 2$, is the water-volume rate increasing? Justify.

Work and justification

EF02 Free response

II A 2 questions total 30 minutes

Graphing calculator required

Let $f(x)=e^{-x^2}+x/4$ and $F(x)=\int_0^x f(t)\,dt$ for $0\le x\le 2$.

(a) Find $F(2)$ and the average value of f on the interval.

Work and justification

(b) Find the absolute maximum value of f and show candidate comparisons.

Work and justification

(c) Justify whether F increases throughout the interval.

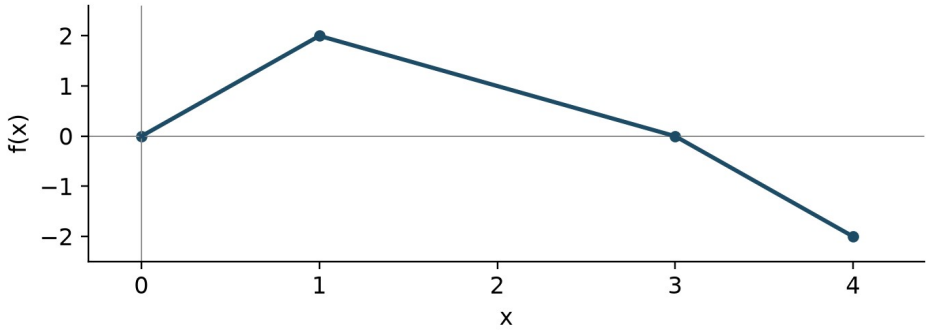
Work and justification

EF03 Free response

II B 4 questions total 60 minutes

Calculator not permitted

The graph shows a continuous f and $G(x)=\int_0^x f(t)\,dt$ on $[0,4]$.



(a) Find $G(4)$ and $G'(2)$.

Work and justification

(b) Justify the absolute maximum of G and its location.

Work and justification

(c) Find all inflection inputs of G and explain why $x=1$ is included.

Work and justification

EF04 Free response

II B 4 questions total 60 minutes

Calculator not permitted

Consider $y' = 2x(y+2)$ with $y(0) = 1$.

(a) Find the initial tangent line and use it to estimate $y(0.1)$.

Work and justification

(b) Find an explicit solution by separation.

Work and justification

(c) Use the second derivative to decide whether the estimate in (a) is high or low.

Work and justification

EF05 Free response

II B 4 questions total 60 minutes

Calculator not permitted

Let R be the region bounded by $y = 4 - x^2$ and the x-axis.

(a) Find the area of R.

Work and justification

(b) Find the volume when R revolves about the x-axis.

Work and justification

(c) A different solid has base R and equilateral-triangle cross sections perpendicular to the x-axis. Find its volume.

Work and justification

EF06 Free response

II B 4 questions total 60 minutes

Calculator not permitted

Let $f(x)=ax+b$ for $x<1$ and $f(x)=x^2+1$ for $x\geq 1$. Choose parameters so that f is differentiable at 1.

(a) Find a, b and state both matching conditions.

Work and justification

(b) Justify MVT on $[0,2]$ and find a point $c\in(0,2)$ satisfying it.

Work and justification

(c) Find the average value of f on $[0,2]$.

Work and justification

Derivative and integral reference

Use each formula on its valid domain. Angles are radians and C is an integration constant. Compositions require an additional chain factor or substitution.

$$(x^n)' = n\,x^{n-1}$$

$$(\ln|x|)' = \frac{1}{x}\,(x \neq 0)$$

$$(a^x)' = a^x \ln a\,(a > 0)$$

$$(\tan x)' = sec^2 x\,,(\cot x)' = -\,csc^2 x$$

$$(sec\,x)' = sec\,x \tan x\,,(csc\,x)' = -\,csc\,x \cot x$$

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}\,,(\arctan x)' = \frac{1}{1+x^2}$$

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C\,(n \neq -1)$$

$$\int \frac{1}{x}\,dx = \ln|x| + C$$

$$\int e^x\,dx = e^x + C\,,\int \cos x\,dx = \sin x + C$$

$$\int \sin x\,dx = -\cos x + C$$

$$\int \frac{1}{1+x^2}\,dx = \arctan x + C$$

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin x + C$$

Bilingual calculus vocabulary

English	한국어
limit / one-sided limit	극한 / 한쪽 극한
continuity / differentiability	연속성 / 미분가능성
derivative / second derivative	도함수 / 둘째 도함수
average rate / instantaneous rate	평균변화율 / 순간변화율
tangent line / linearization	접선 / 선형근사
implicit / inverse function	음함수 / 역함수
critical number / local extremum	임계수 / 극값
absolute maximum and minimum	절대최댓값과 절대최솟값
concavity / inflection point	오목성 / 변곡점
mean value theorem	평균값정리
extreme value theorem	최대최소정리
intermediate value theorem	중간값정리
antiderivative / definite integral	원시함수 / 정적분
accumulation / net change	누적 / 순변화
Riemann sum / trapezoidal sum	리만합 / 사다리꼴합
fundamental theorem of calculus	미적분의 기본정리
slope field / equilibrium solution	기울기장 / 평형해
separation of variables	변수분리
displacement / total distance	변위 / 총 이동거리
cross section / washer	단면 / 와셔

Sources and the BC extension plan

Official framework checked October 28 2026

AP Calculus AB course framework

Open the official College Board source

AP Calculus AB exam

Open the official College Board source

Fall 2026 clarifications

Open the official College Board source

Calculus course and exam description

Open the official College Board source

Sequence for the BC extension

1. Retain the eight AB units and repair diagnostic gaps.

2. Add integration by parts, linear partial fractions, improper integrals, Euler's method, logistic models, and arc length.

3. Write the unit on parametric, vector-valued, and polar functions.

4. Write sequences and series, convergence tests, power series, Taylor polynomials, and error bounds.

5. Publish BC-specific problems and mixed practice exams after separate validation. This package does not contain the BC textbook itself.